Course Code	Name of the Course			
216U03C502 $	**Computer Communication Networks**			
Teaching Scheme	**TH**	**P**	**TUT**	**Total**
	03	**--**	**--**	**03**
Credits Assigned	**03**	**--**	**--**	**03**

Evaluation Scheme	**Marks**				
	LAB/TUT CA	**CA (TH)**		**ESE**	**Total**
		IA	**ISE**		
	--	**20**	**30**	**50**	**100**

216U03C502 $ - The course is common with 216U41C502 of CCE program.

Course pre-requisites:

- Communication Systems
- Basics of Digital communication

Course Objectives: The objective of this course is to enable the students to understand the fundamentals of Computer Networks, OSI layers and TCP/IP model, networking devices. It introduces the concept of IP addressing and subnetting and Routing . The course covers the protocols in various TCP/IP layers such as MAC protocols, ARP, Internet Protocol ,TCP,UDP, DNS, HTTP and Email protocols etc.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Understand layered architecture of communication network, networking devices and the physical layer.

CO2. Explore the significance of data link layer ,error and flow control and MAC protocols

CO3. Understand and apply IP addressing, routing algorithms, and design complete network and its related security as per the given requirements.

CO4. Understand functionalities of transport layer, concept of port numbers and UDP and TCP protocols with their operations

CO5. Understand and demonstrate various application layer protocols as HTTP,DNS,SMTP

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs	CO
1		**Introduction to Data Communication Networks and Physical Layer**	06	CO1
	1.1	Network Criteria, Types of Networks: LAN, WAN, MAN. Concept of Packet switching and Circuit switching.		
	1.2	Need for layered architecture, OSI model and TCP/IP reference model		
	1.3	Networking Devices: Hub, Bridge, Router, Switch		
	1.4	Transmission media: Guided and Unguided media		
2		**The Data Link Layer**	11	CO2
	2.1	Introduction to Data link layer, Frame structure (bit,byte and character stuffing) Error control and Flow control. Error detection : Parity check, Hamming Code CRC and Checksum		
	2.2	Flow control Protocols: Stop-and-wait, Go-Back-N, Selective-Repeat, Piggybacking		
	2.3	Multiple Access Protocols: Controlled Access, Channelization, Random Access protocols: ALOHA, slotted ALOHA, CSMA, CSMA/CD, CSMA/CA.		
	2.4	Concept of Virtual LAN (VLAN), Overview of Ethernet Standards		
3		**The Network Layer**	17	CO3
	3.1	Network layer Services, IPv4 addressing , Subnetting		
	3.2	Internet Protocol (IPv4)-Header format ,Fragmentation in IP		
	3.3	Private and Public IP addresses, Network Address Translation (NAT)		
	3.4	Routing : Distance vector routing, Link State routing. Unicast routing protocols: Overview of RIP , OSPF, BGP		
	3.5	Address Resolution Protocol(ARP), RARP, Dynamic Host Configuration protocol(DHCP), ICMP , IPv6 addressing		
	3.6	Introduction to Network Security: Security Goals, Attacks Services, Techniques Network layer security : IPSec protocol, Two Modes: Tunnel and Transport ,Services Provided by IPSec		
4		**The Transport Layer**	05	CO4
	4.1	Transport Layer services, Connectionless and connection oriented services, Process-to Process connection, concept of port numbers and socket address.		
	4.2	User Datagram Protocol (UDP) ,Transmission Control Protocol (TCP), TCP connection using 3 way Handshaking , TCP Timers		
	4.3	Introduction to SCTP		

5	Application Layer			
	5.1	Application Layer services, HTTP- Hyper Text Transfer Protocol, HTTPs	06	CO5
	5.2	File Transfer Protocol –FTP, Telnet, SCP		
	5.3	Domain Name System-DNS		
	5.4	Email Protocols: SMTP ,PoP3 ,IMAP4		
		Total	**45**	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	B. Forouzan,	*Data Communication and Networking*	McGraw Hill Publication, India	5th Edition, 2013
2	L. Garcia et al	*Communication Networks*	McGraw Hill Publication, India	2nd Edition, 2004
3	J. F. Kurose and K. W. Ross	*Computer Networking: A Top-Down Approach*	Pearson Publication, India	5th Edition, 2012